# Midterm Study Guide

The midterm test is over Learning Unit 1 though 4. This might feel really early in the term, but it is to avoid winter break. If you have been paying attention in class and completing the labs, you are ready for this test, no studying needed. If you have missed class or been avoiding programming the labs yourself, this test will be challenging. Just some advice, sleep is more important than last-minute studying.

## Test Structure

The test is made of two parts. Test will be 60 points total (very roughly 12% of your final grade).

### Multiple Choice, T/F, and fill in the blank questions.

[1pts x 20 questions]

- These questions are reused from the Unit Quiz.
- Coding questions.

### Short Answer/Essay

[8pts x 5 questions]

- This is where your programming skills will shine.
- Each question will give you a scenario and maybe some starter code.
- The code you submit MUST BE VALID JS.
- I will give you points for attempting to solve the problem.
- If you misspell the name of a method or make a small typo, I will be lenient.
- What I am focusing on is your ability to solve the problem and knowledge of the JS language.

> ⚠**THIS IS NOT Pseudocode**. Don't just submit human language.

## Concepts

### LU01 - Into

- L1: Getting Started
  - Syllabus, History, Set up development environment
  - History of JS
  - Setting up students development environment
  - Activity: Visual Check of setup completed
- L2: Where JS
  - Inline JavaScript vs External JavaScript
  - JS in browser
  - JS inline
  - JS internal (block)
  - JS external (`script.js`)
- L3: I/O and Variables
  - `console.log()`

        - `prompt()`
        - `alert()`
        - `confirm()`
    - `let` vs `var` vs `const`
    - `typeof`
    - Have students clone down the lab and run its test suite.
- L4: Conditionals
    - `if`
    - `else`
    - `switch`
    - Math operators and PI

## LU02 - DOM

- L1: Query Selectors
    - DOM diagram
    - Find elements in the DOM
- L2: Changing and Creating Elements
    - Give students an example website and configure different elements on screen.
        - `innerHTML`
        - `innerText`
        - `className`
        - `style`
- L3: Creating Elements
    - Using hard coded data, create new information to put on screen.
        - `createElement()`
        - `appendChild()`
- L4: User Input
    - Use user input (`prompt()`) and display it to screen

## LU03 - Events

- L1: Events
    - `addEventListener()`
    - `onclick` Show but explain why we don't use it.
    - Show off all the different parts of Event Object.
    - Buttons
- L2: Event Listeners
    - Clipboard (Steel your password `inator`)
    - Keyboard (Frustrating text box)
    - Mouse (Mice follow game)
- L3: Forms
    - Getting User Input from form elements
    - When should you validate user input?
        - On submit?
        - On keypress?
    - Prevent Default

- L4: UI Changes
  - User Input → Business Logic → UI Changes

### LU04 – Arrays and Looping

- L1: Arrays
  - Intro to arrays, indexing, `push()`, `pop()`, `length()`
- L2: Looping
  - for, while, for...of
  - Generate dynamic HTML lists (No Template Literals)
- L3: Array Methods
  - forEach, map, filter, find
  - Filtered display of list items (e.g., only even numbers)
- L4: continue and break

## Study Materials

- W3 Schools
- Mozilla
- Video Resources, I really like "Bro Code" here are some of my favorites:
  - JavaScript tutorial for beginners
  - The JavaScript DOM explained in 5 minutes!
  - Learn JavaScript MOUSE EVENTS in 10 minutes!
  - Learn JavaScript ARRAYS in 8 minutes!
  - Learn JavaScript FOR LOOPS in 5 minutes!